

Internship Project Phase 2 Report (BUILD)

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Project: E-Commerce Sales Performance Dashboard with Cohort Analysis

External guide:

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1. Introduction

Phase 2 focuses on transforming raw transactional data into meaningful insights using Python. This phase involves data cleaning, exploratory data analysis (EDA), and cohort analysis to understand customer behavior and sales performance.

2. Objectives

- To clean and preprocess the dataset for accurate analysis
- To perform exploratory data analysis to identify trends and patterns
- To analyze customer behavior using cohort analysis
- To generate insights for business decision-making
- To prepare data for dashboard visualization in Power BI

3. Dataset Description

The dataset used is the **Online Retail II dataset**, which contains transactional data of a UK-based online retail store.

Key attributes include:

- Invoice: Unique transaction ID
- StockCode: Product code
- Description: Product name
- Quantity: Number of items purchased
- InvoiceDate: Date and time of transaction
- Price: Price per unit
- Customer ID: Unique customer identifier
- Country: Customer location

4. Data Cleaning and Preprocessing

Data cleaning was performed to ensure accuracy and reliability of analysis.

Steps performed:

- Removed rows with missing Customer IDs
- Removed cancelled transactions (Invoice starting with 'C')
- Removed negative and zero quantities
- Removed invalid price values ($\text{Price} \leq 0$)
- Handled outliers in quantity
- Converted InvoiceDate to datetime format
- Created a new column: **Revenue = Quantity × Price**

After applying all cleaning steps, the dataset was reduced from **5,41,910** to **3,97,883** valid transactions, retaining approximately **73%** of the original data.

5. Feature Engineering

New features were created to enable time-based analysis:

- InvoiceMonth → Monthly grouping of transactions
- Year → Year of transaction
- Month → Month number

These features helped in analyzing trends and customer behavior over time.

6. Exploratory Data Analysis (EDA)

EDA was performed to understand sales performance and customer patterns.

6.1 Monthly Revenue Trend

Insight:

Revenue shows a strong increasing trend with a peak towards the end of the year, indicating high seasonal demand during the holiday period.

Recommendation:

The business should prepare for increased demand by stocking inventory in advance and launching promotional campaigns starting from October to maximize revenue during peak seasons.

6.2 Top Products Analysis

Insight:

A small number of products contribute significantly to total revenue, indicating a highly skewed sales distribution.

Recommendation:

The business should prioritize these top-performing products for promotions, ensure consistent stock availability, and use them in marketing campaigns to drive higher sales.

6.3 Country-wise Sales Analysis

Insight:

The United Kingdom dominates sales, indicating a strong domestic market presence.

Recommendation:

The business should strengthen its position in the UK while also expanding marketing efforts in other high-performing countries such as Netherlands and Germany to diversify revenue sources.

6.4 Orders and Customer Growth

Insight:

Both the number of orders and unique customers show a consistent upward trend, indicating business growth.

Recommendation:

The business should maintain this growth by implementing customer engagement strategies such as loyalty programs, personalized offers, and retention campaigns.

6.5 Revenue Distribution

Insight:

Revenue distribution is highly skewed, with most transactions being low value and a few high-value purchases contributing significantly.

Recommendation:

The business should focus on increasing average order value by offering product bundles, discounts on bulk purchases, and upselling strategies.

7. Cohort Analysis

Key Findings:

- Customer retention is highest during the first month after purchase
- Retention decreases significantly in subsequent months

- Some cohorts show relatively better retention, indicating stronger customer engagement during specific periods

Recommendation:

The business should focus on improving retention immediately after the first purchase by implementing follow-up strategies such as email reminders, discount offers, and loyalty incentives to encourage repeat purchases.

7.1 Methodology

- Identified first purchase month for each customer
- Calculated time difference between purchases (Cohort Index)
- Created a retention matrix
- Visualized retention using a heatmap

7.2 Key Findings

- Customer retention is highest during the first month
- Retention decreases gradually over time
- Some cohorts show better retention, indicating strong customer engagement during certain periods

8. Output Generated

- Cleaned dataset (final_cleaned_data.csv)
- Cohort retention dataset (cohort_retention.csv)
- Visualizations for trends and patterns

These outputs were used for dashboard creation in the next phase.

9. Conclusion

Phase 2 successfully transformed raw data into actionable insights. The analysis revealed strong sales growth, seasonal trends, product concentration, and customer retention patterns. These findings provide valuable input for business decision-making and dashboard visualization.